Model Performance Summary

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CIFAR-10 Image Classification using Deep Learning — Master's Research Project

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1. Objective

Build, evaluate and compare four convolutional architectures on CIFAR-10 (60,000 colour images, 10 balanced classes) — one custom CNN trained from scratch and three transfer-learning models with frozen ImageNet backbones — and determine which is best suited to production deployment under real compute constraints.

2. Method at a glance

Aspect	Setting
Data source	official CIFAR-10 archive → `train.csv` (50,000 × 3,073), `test.csv` (10,000 × 3,073)
Split	40,000 train / 10,000 validation / 10,000 test (test never seen during training)
Input	32×32 native, upsampled in-graph to 224×224×3
Normalisation	÷255 → [0,1]; per-channel standardisation from the **training subset only**
Augmentation	rotation ±15°, shift ±10%, zoom ±10%, horizontal flip — **training partition only**
Models	Simple CNN (scratch); VGG16 / ResNet50 / EfficientNetB0 (frozen backbone + GAP→Dense(128)→Dropout(0.5)→Dense(10))
Optimizer / loss	Adam (lr 0.001) / categorical cross-entropy
Batch / epochs	32 / up to 15
Callbacks	EarlyStopping(val_accuracy, patience 5, restore best); ReduceLROnPlateau(val_loss, factor 0.2, patience 3)
Seed	42 (Python, NumPy, TensorFlow, PYTHONHASHSEED)

3. Headline results — 10,000 unseen test images

Model	Test Acc	F1 (w)	F1 (macro)	Top-5	Params	Trainable	Epochs	Train (min)	ms/img	Size (MB)
ResNet50_Transfer	0.9056	0.9051	0.9051	0.9976	23,851,274	263,562	15	94.5	7.409	91.0
EfficientNetB0_Transfer	0.8974	0.8971	0.8971	0.9970	4,214,829	165,258	15	58.8	4.444	16.1
VGG16_Transfer	0.8441	0.8429	0.8429	0.9942	14,781,642	66,954	15	143.8	9.813	56.4
Simple_CNN	0.5533	0.5476	0.5476	0.9472	12,939,722	12,939,722	15	44.0	0.922	49.4
Simple_CNN_Native32	0.7315	0.7229	0.7229	0.9798	356,810	356,810	15	5.6	0.115	1.4

Best-performing model: ResNet50_Transfer — 0.9056 test accuracy, 0.9051 weighted F1, +0.3523 above the weakest architecture. McNemar's exact test on paired predictions confirms the leading differences are statistically significant rather than sampling noise.

4. Training behaviour

Model	Epochs run	Early stopped	Best epoch	Best val acc	LR start	LR end	LR reductions	Train-val gap
ResNet50_Transfer	15	no	15	0.9136	1.0e-03	2.0e-04	1	-0.0588
EfficientNetB0_Transfer	15	no	15	0.9024	1.0e-03	1.0e-03	0	-0.0812
VGG16_Transfer	15	no	14	0.8467	1.0e-03	2.0e-04	1	-0.0637
Simple_CNN	15	no	13	0.5557	1.0e-03	4.0e-05	2	-0.1086
Simple_CNN_Native32	15	no	13	0.7363	1.0e-03	2.0e-04	1	-0.0550

5. Feature analysis

Model	Layer	Map	Raw dims	Pooled	Sparsity %	Act mean	Act std	Silhouette	2-D 10-NN	Linear probe
ResNet50_Transfer	conv5_block3_out	7×7×2048	100,352	2,048	83.33	0.472	1.765	0.0953	0.7610	0.8475
EfficientNetB0_Transfer	top_activation	7×7×1280	62,720	1,280	0.00	0.090	0.823	0.1361	0.7720	0.8525
VGG16_Transfer	block5_pool	7×7×512	25,088	512	87.61	1.221	4.866	-0.0235	0.6095	0.7515
Simple_CNN	conv3_128	56×56×128	401,408	128	88.42	0.232	1.082	-0.1339	0.2515	0.4485

6. Advanced experiments

Experiment	Result
Fine-tuning (ResNet50_Transfer, last stage, lr 1e-5)	0.9056 → 0.9334 (**+0.0278**)
Cross-validation (3-fold, Simple CNN)	0.4228 ± 0.0084 validation accuracy
Best ensemble (`top2_soft_voting`)	0.9219 (+0.0163 vs best single model)
Compression	TFLite float16 / dynamic int8 + dense-head pruning — see Analysis Report §7
Lowest batch-1 latency	Simple_CNN at 3.47 ms (p50)

7. Key findings

1. **ResNet50_Transfer is the most accurate architecture** at 0.9056, and the ranking is statistically significant, not noise.
2. **Parameter count predicts training cost backwards.** ResNet50 carries 1.6× VGG16's parameters (23.9 M vs 14.8 M) and updates 4× as many (264 K vs 67 K), yet trains in roughly two-thirds of VGG16's wall-clock time *and* scores higher. VGG16 stacks wide 3×3 convolutions at high spatial resolution; ResNet50 spends most of its depth on 1×1 bottlenecks over reduced feature maps, doing far less arithmetic per forward pass despite holding more weights. Freezing removes the backward pass, but it cannot make an arithmetic-heavy backbone cheap.
3. **Learning rate is the highest-leverage hyper-parameter.** One order of magnitude in either direction moved accuracy more than any architecture choice. `ReduceLROnPlateau` demonstrably rescued the Simple CNN's headline run, which was diverging by epoch 3.
4. **Deeper can mean smaller.** Adding a `Conv2D(256)` block to the Simple CNN halved the `Flatten` vector and *reduced* total parameters — 99 % of that architecture's weights live in the `Flatten → Dense` transition, not in its convolutions.
5. **Upsampling adds pixels, not information — and for a from-scratch network it is actively harmful.** The identical Simple CNN architecture fed at CIFAR's native 32×32 uses 36× fewer parameters, trains 7.9× faster, infers 8.0× faster, and is **+0.1782 more accurate** (0.7315 vs 0.5533). Resizing cannot create information the 32×32 source never held; it only inflates `Flatten → Dense(128)` to 12.85 M parameters and destabilises training. 224 is the right input *only* for the pre-trained backbones.
6. **Feature dimensionality is not feature quality.** The Simple CNN's last convolutional layer has the largest raw feature map yet is among the weakest by linear probe and t-SNE separability.

8. Production recommendation

- **Maximum accuracy** — ResNet50_Transfer, optionally fine-tuned (Analysis Report §7.1, worth +0.0278).
- **Real-time / mobile** — Simple_CNN, lowest batch-1 latency at 3.47 ms (≈ 268 fps), combined with float16 TFLite quantisation for the size reduction measured in §7.4.
- **Constrained retraining** — the frozen transfer models converge within a few epochs and are markedly more robust to hyper-parameter error than the from-scratch CNN.

9. Limitations

The 32×32 source resolution caps every result; frozen backbones leave measurable headroom (quantified by the fine-tuning experiment); the parameter-modification study uses a reduced 8,000-image / 6-epoch protocol whose **rankings are controlled but whose absolute values are not comparable** to the headline numbers; and outside the cross-validation there is one run per configuration, so differences below ≈ 0.0167 should not be treated as meaningful without a significance test.

Full methodology, all 15 required question answers, figures and per-class results are in `Analysis_Report.md` *and* `CIFAR10_CNN_Capstone.ipynb`*.*